

03 Rolling Validation Selection Error Rate (Expanding Window)

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Goal (expanding window)

One-step-ahead **expanding** rolling validation: initial training window $w = \text{floor}(0.5 * n)$, then for $t = w, \dots, n - 1$ fit on rows $1:t$ and predict row $t + 1$. Mean squared prediction error is averaged over $n - w$ steps. We compare a **linear** model (M1) and **degree-3 polynomial** (M3), select the model with lower mean RV MSE (tie $\rightarrow$ linear), and report the Monte Carlo **selection error rate** vs n on 50, 100, ..., 1000 (step 50), with 1,000 replications per n .

Linear truth

Simulation settings

Table 1: Rolling validation settings (expanding window, linear truth).

setting	value
seed	2026
window type	expanding
train_frac ($w = \text{floor}(\text{train_frac} * n)$)	0.5
horizon	1
Monte Carlo replications per n	1000
Sample sizes (grid)	50, 100, ..., 1000 (step 50)
beta0	1
beta1	2
sigma	4
poly_degree	3
truth	linear: $y = \text{beta0} + \text{beta1} * x + \text{eps}$

setting	value
selection rule	Choose model with lower mean RV MSE; tie -> linear

Error rate summary

Table 2: Monte Carlo mean selection-error rate (truth is linear; error = poly chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.011	0.00330
100	0.011	0.00330
150	0.008	0.00282
200	0.008	0.00282
250	0.009	0.00299
300	0.008	0.00282
350	0.012	0.00344
400	0.005	0.00223
450	0.014	0.00372
500	0.008	0.00282
550	0.013	0.00358
600	0.011	0.00330
650	0.005	0.00223
700	0.012	0.00344
750	0.007	0.00264
800	0.012	0.00344
850	0.013	0.00358
900	0.011	0.00330
950	0.008	0.00282
1000	0.013	0.00358

Error rate vs sample size

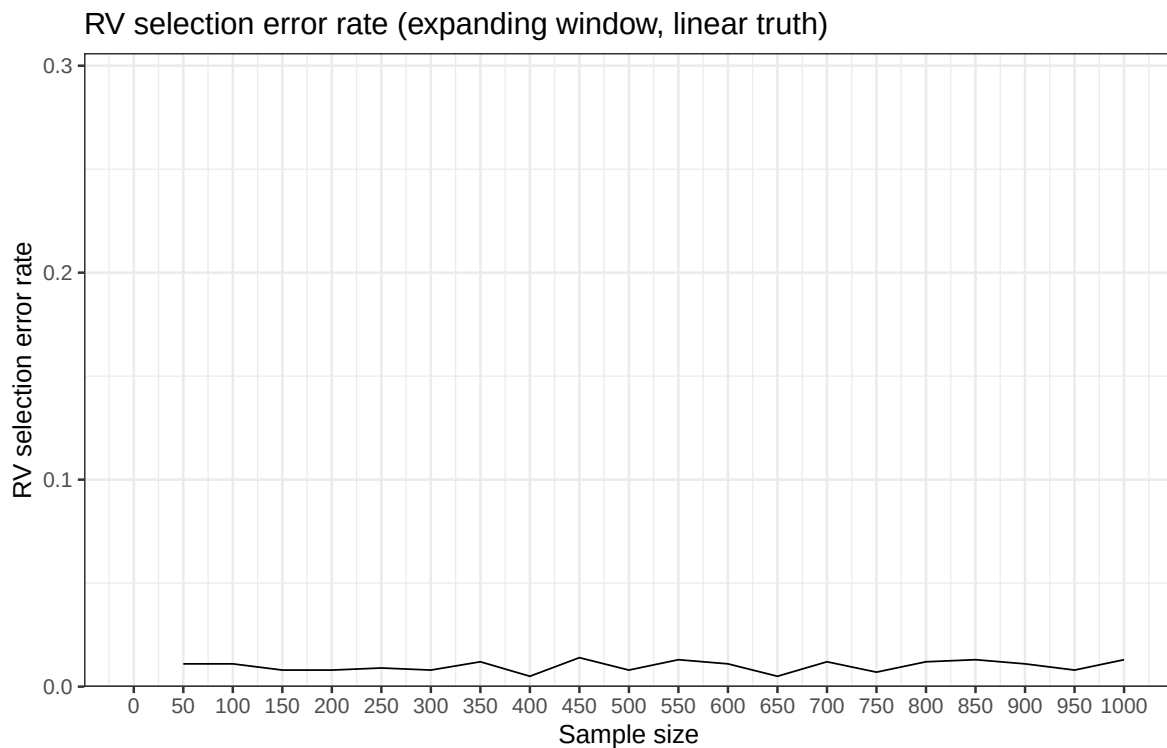


Figure 1: RV selection error rate under linear truth (expanding window).

Polynomial truth

Simulation settings

Table 3: Rolling validation settings (expanding window, polynomial truth).

setting	value
seed	2026
window type	expanding
train_frac (w = floor(train_frac * n))	0.5
horizon	1
Monte Carlo replications per n	1000
Sample sizes (grid)	50, 100, ..., 1000 (step 50)

setting	value
beta0	1
beta1	2
beta2	1
beta3	1
sigma	4
poly_degree	3
truth	polynomial: $y = \text{beta0} + \text{beta1}x + \text{beta2}x^2 + \text{beta3}x^3 + \text{eps}$
selection rule	Choose model with lower mean RV MSE; tie -> linear

Error rate summary

Table 4: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.427	0.01564
100	0.096	0.00932
150	0.022	0.00464
200	0.001	0.00100
250	0.000	0.00000
300	0.000	0.00000
350	0.000	0.00000
400	0.000	0.00000
450	0.000	0.00000
500	0.000	0.00000
550	0.000	0.00000
600	0.000	0.00000
650	0.000	0.00000
700	0.000	0.00000
750	0.000	0.00000
800	0.000	0.00000
850	0.000	0.00000
900	0.000	0.00000
950	0.000	0.00000
1000	0.000	0.00000

Error rate vs sample size

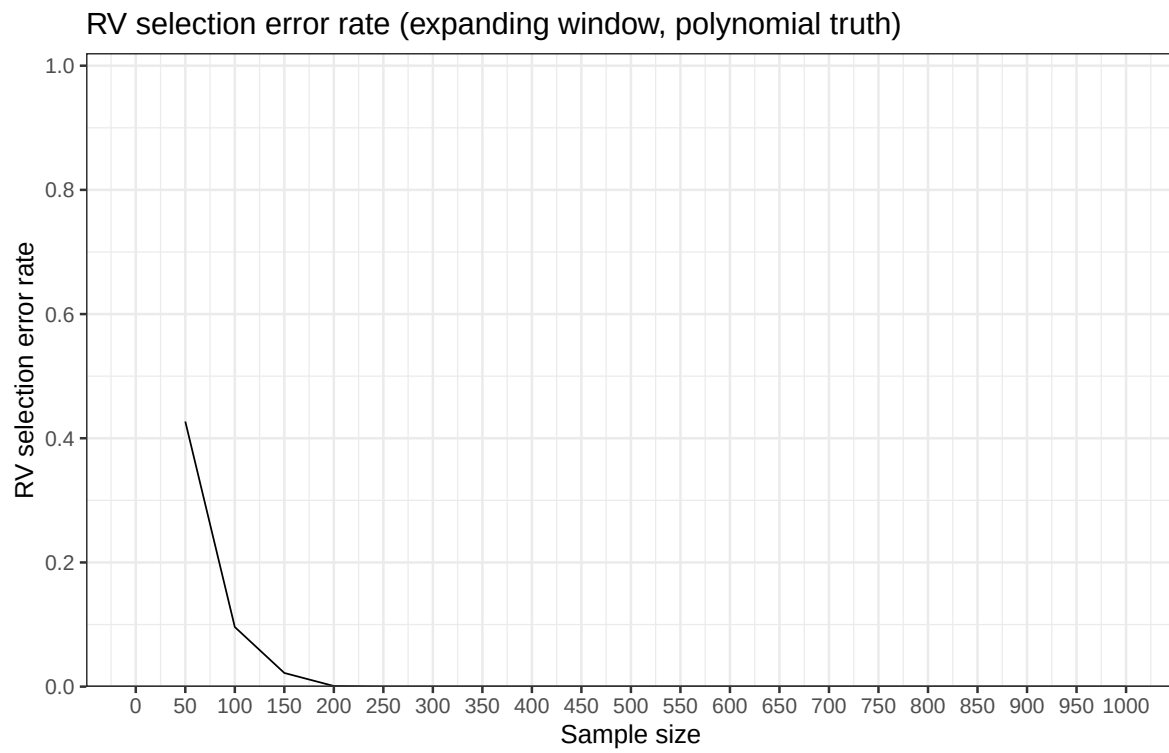


Figure 2: RV selection error rate under polynomial truth (expanding window).